Optimal Stopping for Perpetual American Call Option

Kenneth Anderson, Chen Yao

July 12, 2026

SLMath Summer School

Dynamical Systems for Machine Learning and AI

Hosted by SLMath and IBM Research





SIMONS LAUFER
MATHEMATICAL
SCIENCES INSTITUTE

Roadmap

1. Recall the optimal stopping formulation.
2. Derive the obstacle problem for a perpetual American call.
3. Solve the continuation ODE by a power ansatz.
4. Determine the exercise boundary by value matching and smooth fit.
5. Interpret the no-dividend benchmark.

Optimal Stopping Framework

General Formulation of Optimal Stopping

Let X denote the underlying asset price evolving as the diffusion

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t,$$

with infinitesimal generator

$$\mathcal{L}\phi(x) = b(x)\phi'(x) + \frac{1}{2}\sigma^2(x)\phi''(x).$$

The holder chooses an admissible exercise time from

$$\mathcal{T} = \{\tau : \tau \text{ is an } (\mathcal{F}_t)\text{-stopping time with values in } [0, \infty]\},$$

where f is the running reward rate, g is the payoff received upon exercise, and $\rho > 0$ is the discount rate. The value function is

$$v(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x \left[\int_0^\tau e^{-\rho t} f(X_t)dt + e^{-\rho\tau} g(X_\tau) \right].$$

Dynamic Programming Principle

For any stopping time θ, the DPP gives

$$v(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x \left[\int_0^{\tau \wedge \theta} e^{-\rho t} f(X_t) dt + e^{-\rho \tau} g(X_\tau) \mathbf{1}_{\{\tau < \theta\}} + e^{-\rho \theta} v(X_\theta) \mathbf{1}_{\{\theta \leq \tau\}} \right].$$

Two immediate implications:

$$v(x) \geq g(x), \qquad \text{because } \tau = 0 \text{ is admissible.}$$

If we choose $\theta = h$ for small $h > 0$, then

$$v(x) \geq \mathbb{E}_x \left[\int_0^h e^{-\rho t} f(X_t) dt + e^{-\rho h} v(X_h) \right].$$

Deriving the Continuation Equation

Assume v is C^2 on a region where continuation is optimal. Applying Itô's formula to $e^{-\rho t}v(X_t)$ yields

$$d(e^{-\rho t}v(X_t)) = e^{-\rho t}(\mathcal{L}v - \rho v)(X_t)dt + e^{-\rho t}v'(X_t)\sigma(X_t)dW_t.$$

If we do not stop before a short time h, the DPP is locally an equality:

$$v(x) = \mathbb{E}_x\left[\int_0^h e^{-\rho t}f(X_t)dt + e^{-\rho h}v(X_h)\right].$$

Plugging the Itô expansion of $e^{-\rho t}v(X_t)$ into this identity gives

$$0 = \mathbb{E}_x\left[\int_0^h e^{-\rho t}(f + \mathcal{L}v - \rho v)(X_t)dt\right].$$

Continuation PDE and Obstacle Problem

Dividing by h and letting $h \downarrow 0$, we obtain the continuation Hamiltonian equation

$$\rho v - \mathcal{L} v - f = 0.$$

Since the American option allows immediate exercise, there are two local possibilities:

$$\begin{array}{rcll} \text{continue} & : & v(x) > g(x), & \rho v - \mathcal{L} v - f = 0, \\ \text{stop} & : & v(x) = g(x). & \end{array}$$

The value must dominate the exercise payoff, while the continuation equation holds only where waiting is optimal. Combining these alternatives gives the variational inequality

$$\min\{\rho v - \mathcal{L} v - f,\ v - g\} = 0.$$

Stopping and Continuation Regions

Define

$$\mathcal{C} = \{x : v(x) > g(x)\}, \qquad \mathcal{S} = \{x : v(x) = g(x)\}.$$

Region	Meaning	Equation
$\mathcal{C}$	Continue	$\rho v - \mathcal{L}v - f = 0$
$\mathcal{S}$	Stop	$v = g$
$\partial\mathcal{C}$	Free boundary	Value matching and smooth fit

The candidate optimal time is usually the first hitting time of the stopping region:

$$\tau^* = \inf\{t \geq 0 : X_t \in \mathcal{S}\}.$$

Perpetual American Call Option

Problem: Perpetual American Call

Assume the underlying asset price X follows a geometric Brownian motion

$$dX_t = (r - \delta)X_t dt + \sigma X_t dW_t, \qquad X_0 = x > 0,$$

where r is the risk-free interest rate, δ is the continuous dividend yield, and σ is the stock volatility.

A perpetual American call with strike K has value

$$v(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x \left[e^{-r\tau} (X_\tau - K)^+ \right].$$

Compared to the general optimal stopping formulation, we have

$$f \equiv 0, \qquad g(x) = (x - K)^+, \qquad \rho = r.$$

The holder can buy the asset for K at any exercise time. The central question is: **when should we exercise, and when should we wait?**

Why an Upper Exercise Threshold?

Assume $\delta > 0$. We expect a stopping set of the form

$$\mathcal{S} = [x^*, \infty), \qquad \mathcal{C} = (0, x^*), \qquad x^* > K.$$

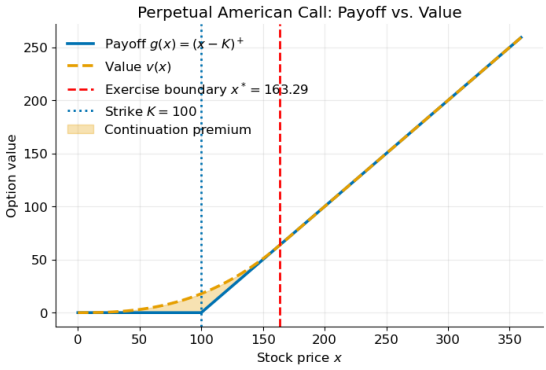
Why is this natural?

- If $x \leq K$, immediate exercise gives 0, while waiting keeps the possibility that X_t later moves above K.
- The payoff $g(x) = (x - K)^+$ is increasing in x: higher stock prices make immediate exercise more attractive.
- When $\delta > 0$, waiting means not holding the stock and forgoing dividend flow; this opportunity cost grows with x.
- For GBM, higher initial prices scale the future path upward: $X_t^{x_2} = (x_2/x_1)X_t^{x_1}$ for $x_2 > x_1$.

Thus the stopping region is expected to be an upper set, leading to

$$\tau^* = \inf\{t \geq 0 : X_t \geq x^*\}.$$

Free-Boundary Picture



$$\mathcal{C} = (0, x^*), \qquad \mathcal{S} = [x^*, \infty).$$

On $\mathcal{S}$:

$$v(x) = x - K.$$

On $\mathcal{C}$, solve the continuation ODE:

$$rv - \mathcal{L}v = 0.$$

At x^*, impose value matching and smooth fit.

DPP and Variational Inequality

The generator of X is

$$\mathcal{L}\phi(x) = (r-\delta)x\phi'(x) + \frac{1}{2}\sigma^2 x^2 \phi''(x).$$

The obstacle problem is

$$\min\left\{rv - (r-\delta)xv' - \frac{1}{2}\sigma^2 x^2 v'',\ v - (x-K)^+\right\} = 0.$$

In the continuation region $(0, x^*)$,

$$rv - (r-\delta)xv' - \frac{1}{2}\sigma^2 x^2 v'' = 0.$$

The free-boundary problem is to determine both v and the unknown threshold x^*.

Continuation ODE: Power Ansatz

On $\mathcal{C} = (0, x^*)$, try a power solution

$$v(x) = x^m.$$

Substitution into

$$rv - (r-\delta)xv' - \frac{1}{2}\sigma^2 x^2 v'' = 0$$

gives the characteristic equation

$$q(m) := \frac{1}{2}\sigma^2 m(m-1) + (r-\delta)m - r = 0.$$

The two roots are

$$m_\pm = \frac{1}{2} - \frac{r-\delta}{\sigma^2} \pm \sqrt{\left(\frac{r-\delta}{\sigma^2} - \frac{1}{2}\right)^2 + \frac{2r}{\sigma^2}},$$

and we have

- $m_- < 0$ directly from the formula.
- $\beta := m_+ > 1$, because $q(1) = -\delta < 0$ and $q(m) \to +\infty$ as $m \to +\infty$.

Continuation Solution

The general continuation solution is

$$v(x) = Ax^{\beta} + Bx^{m_-}, \qquad 0 < x < x^*.$$

The negative-power term explodes as $x \downarrow 0$:

$$x^{m_-} \to \infty.$$

A call option is worthless at zero stock price, so we impose the natural boundary behavior

$$v(0+) = 0.$$

Indeed, if $X_0 = 0$, then $X_t = 0$ for all t, so $v(0) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_0[e^{-r\tau}(X_\tau - K)^+] = 0$.

Therefore,

$$B = 0, \qquad v(x) = Ax^{\beta}, \qquad 0 < x < x^*.$$

Boundary Conditions at x^*

In the stopping region, $v(x) = x - K$ for $x \geq x^*$.

Value matching gives

$$A(x^*)^\beta = x^* - K.$$

Smooth fit gives

$$A\beta(x^*)^{\beta-1} = 1.$$

Dividing smooth fit by value matching:

$$\frac{\beta}{x^*} = \frac{1}{x^* - K}.$$

Hence

$$x^* = \frac{\beta}{\beta - 1}K \quad \text{and} \quad A = \frac{x^* - K}{(x^*)^\beta}.$$

Final Value Function and Optimal Rule

The perpetual American call value is

$$v(x) = \begin{cases} (x^* - K)\left(\frac{x}{x^*}\right)^{\beta}, & 0 < x < x^*, \\ x - K, & x \geq x^*, \end{cases}$$

where

$$\beta = \frac{1}{2} - \frac{r-\delta}{\sigma^2} + \sqrt{\left(\frac{r-\delta}{\sigma^2} - \frac{1}{2}\right)^2 + \frac{2r}{\sigma^2}}, \qquad x^* = \frac{\beta}{\beta - 1} K.$$

The optimal exercise time is

$$\tau^* = \inf\{t \geq 0 : X_t \geq x^*\}.$$

No-Dividend Benchmark

If $\delta = 0$, then $\beta = 1$, so

$$x^* = \frac{\beta}{\beta - 1} K = +\infty \quad \text{and} \quad \mathcal{C} = (0, +\infty).$$

Thus there is no finite optimal exercise boundary for the perpetual call.

Under

$$dX_t = rX_t dt + \sigma X_t dW_t,$$

$e^{-rt} X_t$ is a martingale. For $\tau_H = \inf\{t : X_t \geq H\}$, continuity gives $X_{\tau_H} = H$, and hence

$$x = \mathbb{E}_x \left[e^{-r\tau_H} X_{\tau_H} \right] = H \mathbb{E}_x \left[e^{-r\tau_H} \right] \quad \Longrightarrow \quad \mathbb{E}_x \left[e^{-r\tau_H} \right] = \frac{x}{H}.$$

$$\mathbb{E}_x \left[e^{-r\tau_H} (X_{\tau_H} - K) \right] = (H - K) \mathbb{E}_x \left[e^{-r\tau_H} \right] = (H - K) \left(\frac{x}{H} \right).$$

Sending $H \to \infty$ yields

$$v(x) = x,$$

but the supremum is not attained at any finite threshold.

Verification Idea

Let v be the candidate value function. Then:

$$rv - \mathcal{L}v = 0, \qquad v > (x-K)^+, \qquad 0 < x < x^*.$$

On the stopping region:

$$v = x - K, \qquad rv - \mathcal{L}v = \delta x - rK \geq 0, \qquad x \geq x^*.$$

Hence

$$\min\{rv - \mathcal{L}v,\ v - (x-K)^+\} = 0.$$

Probabilistically, $e^{-rt}v(X_t)$ is a supermartingale and becomes a martingale before τ^*. Thus

$$v(x) = \mathbb{E}_x\left[e^{-r\tau^*}(X_{\tau^*} - K)^+\right] = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x\left[e^{-r\tau}(X_\tau - K)^+\right].$$

Comparative Statics and Takeaways

- The perpetual call problem is an optimal stopping problem with an upper free boundary.
- A positive dividend yield $\delta > 0$ creates a cost of waiting and makes early exercise optimal above x^*.
- The continuation value has the power form Ax^β, where $\beta > 1$ is determined by the characteristic equation.
- Value matching and smooth fit determine the exercise boundary:

$$x^* = \frac{\beta}{\beta - 1} K \quad \text{and} \quad A = \frac{x^* - K}{(x^*)^\beta}.$$

- In the no-dividend case $\delta = 0$, the boundary moves to infinity, so there is no finite optimal exercise threshold.

References


H. Pham.
Continuous-time Stochastic Control and Optimization with Financial Applications.
Springer, 2009.


S. E. Shreve.
Stochastic Calculus for Finance II: Continuous-Time Models.
Springer, 2004.

Acknowledgements

We gratefully acknowledge SLMath, IBM Research, and all instructors and TAs for organizing and supporting this summer school.





SIMONS LAUFER
MATHEMATICAL
SCIENCES INSTITUTE